

Preparing High Fidelity Thermofield Double States

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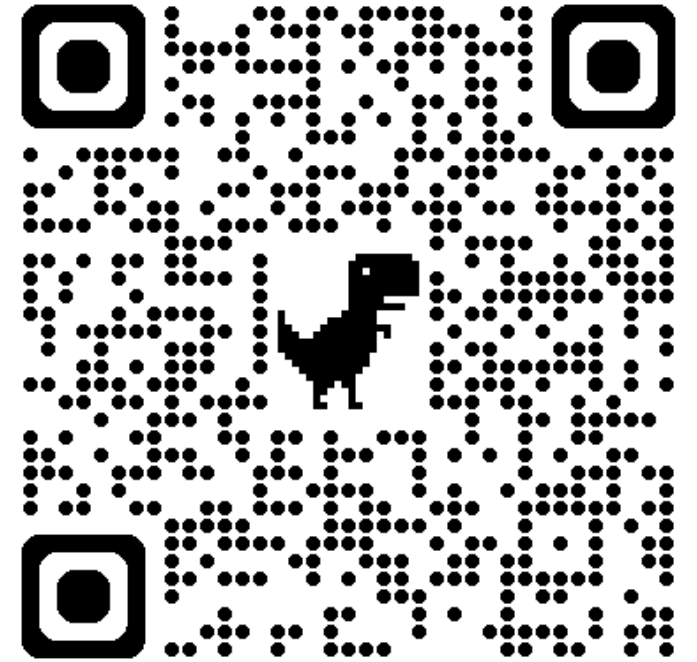
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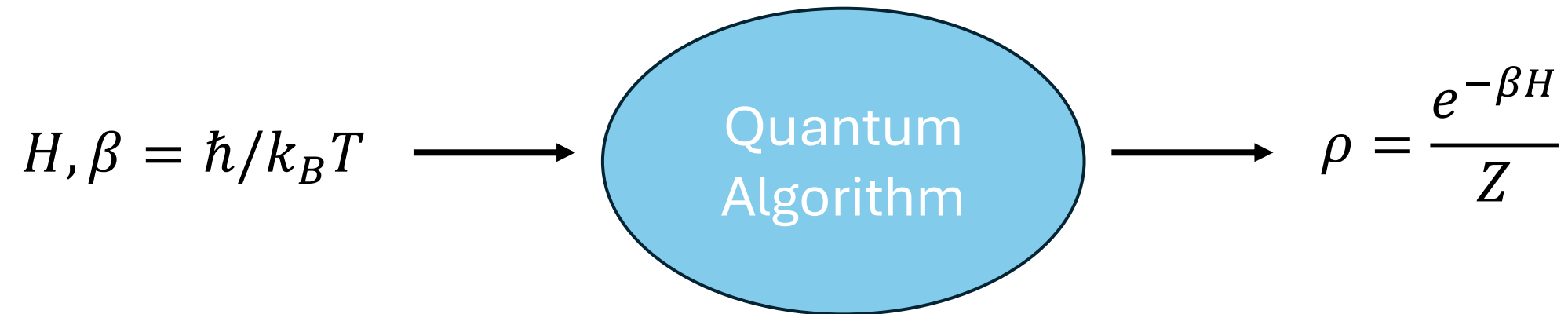


Thermal equilibrium

- Many physical systems can be idealized as existing in thermal equilibrium; local thermal equilibrium is even more common
- If we could prepare thermal states efficiently on a quantum computer, we could do a lot of interesting work
- For example:
 - We could study basic science problems, e.g. high temp superconductors, spin liquids, quark gluon plasma, holographic black holes ...
 - We could search for novel materials with desirable electronic, magnetic, etc. properties
 - We might open the door to novel optimization algorithms, e.g. by recasting optimization problem as an energy minimization

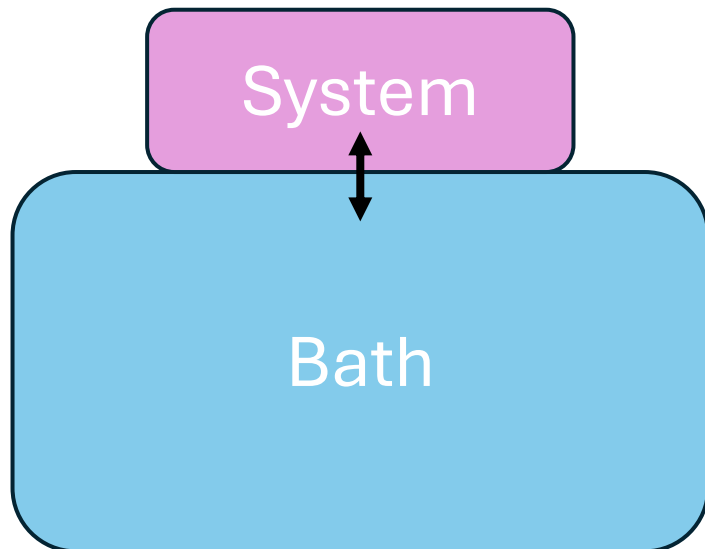
Preparing thermal states

- Given the wide-ranging nature of these problems, we naturally want to have **general purpose algorithms** to efficiently prepare thermal states



Is this even possible?

- Nature provides us with one relatively general algorithm: couple the system of interest to a heat bath at the desired temperature and let the system come into equilibrium with the bath
- This is general purpose but it may take a long time to equilibrate*



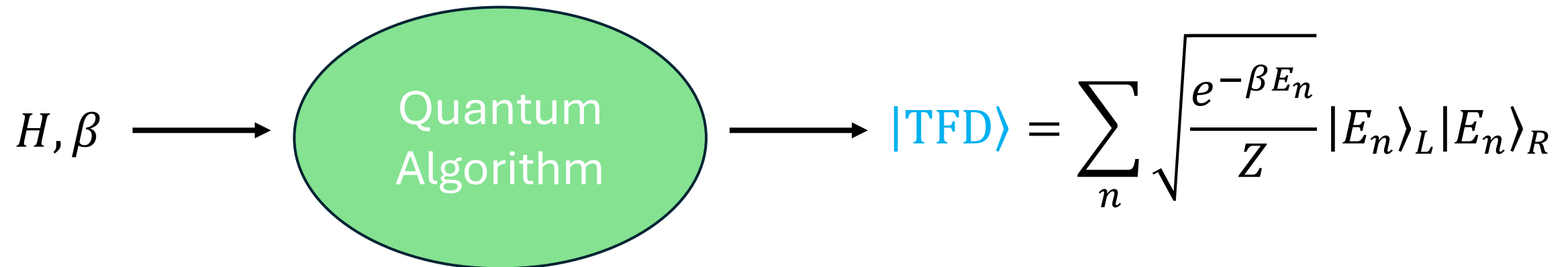
$$\rho_S(t \rightarrow \infty) \approx e^{-\beta H} / Z$$

*This is a real limitation, for example:

- Glasses may not equilibrate on any reasonable timescale
- Critical slowing down at a continuous phase transition
- Encoding of hard optimization problems suggests a barrier

Thermofield double states

- Sometimes we want something stronger: a special structured purification of the thermal state, a **thermofield double state**
- This state shows up in quantum gravity, field theory, and many-body physics in various important ways
- **Nature's algorithm does NOT provide this**



What is known?

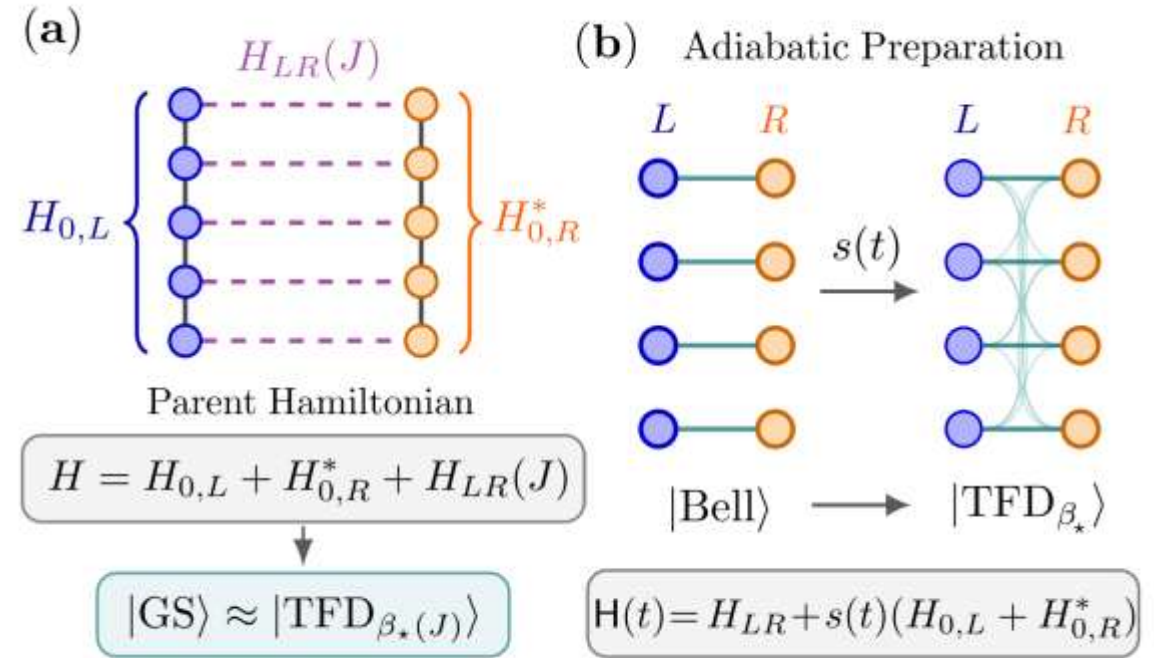
For mixed thermal states, there are a variety of approaches:

- Open system dynamics
- Sampling methods (e.g. [Chen et al. '23](#), $O(\beta)$ footprint Lindblads)
- Variational methods
- Adiabatic methods
- “Imaginary” time methods
- (whole host of classical methods as well)

Some of these can be adapted to address the TFD

Our protocol

$$H = H_{0,L} + H_{0,R}^* + \frac{JN}{2K} \sum_{\alpha=1}^K (\mathcal{O}_{\alpha,L} - \mathcal{O}_{\alpha,R}^*)^2 ,$$



- Part 1: Construct a Hamiltonian family whose GS is an approximate TFD at a tunable temp (**controlled by LR coupling**), adiabatic preparation
- Part 2: Improved fidelity by adding a penalty term that favors a diagonal subspace, $|E_n\rangle_L |E_n\rangle_R$

We provide evidence that this will work when the system self-thermalizes

Background for our approach

- There are a few early works e.g. Feiguin-Klich '13 and McGreevy-S '16 that looked for parent Hamiltonians for TFD states
- A breakthrough came with Maldacena-Qi '18 (MQ) who proposed a simple approach for SYK and studied its gravity dual
- Cottrell et al. '18 suggested that MQ generalized to systems obeying eigenstate thermalization, but some issues emerged
- Bentsen-Nguyen-S '23 and Schuster-Kobrin-Su-Marrochio-Yao '25 discussed an adiabatic approach to state preparation in SYK
- In our paper, we show concretely how to generalize MQ to spins etc., clarify the issues in Cottrell et al., and provide solutions

Example: mixed field Ising chain

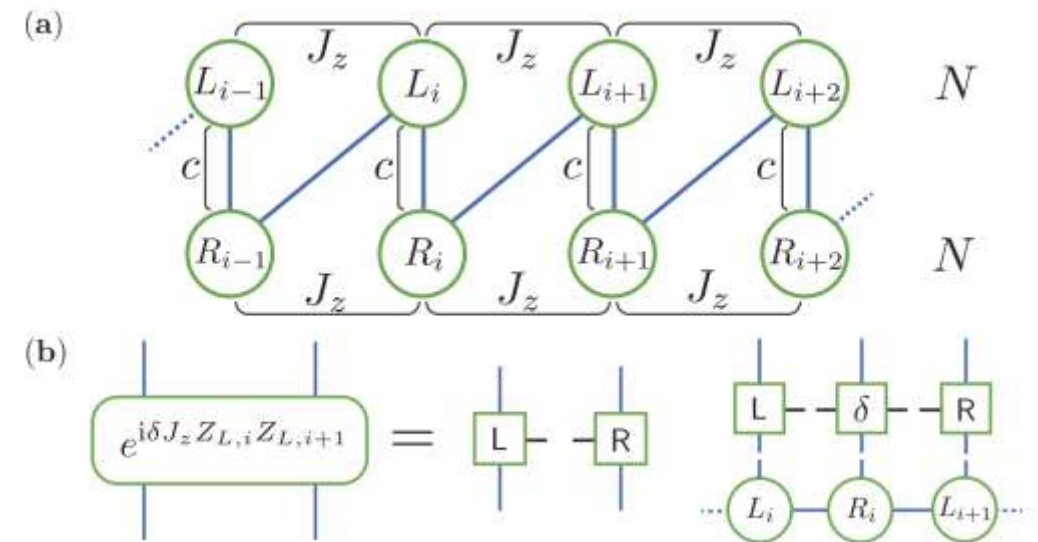
$$H_{\text{MFI},L/R} = J_z \sum_{i=1}^{N-1} Z_{i,L/R} Z_{i+1,L/R} + h_x \sum_{i=1}^N X_{i,L/R}$$

$$+ h_z \sum_{i=1}^N Z_{i,L/R},$$

$$H = H_{\text{MFI},L} + H_{\text{MFI},R}^*$$

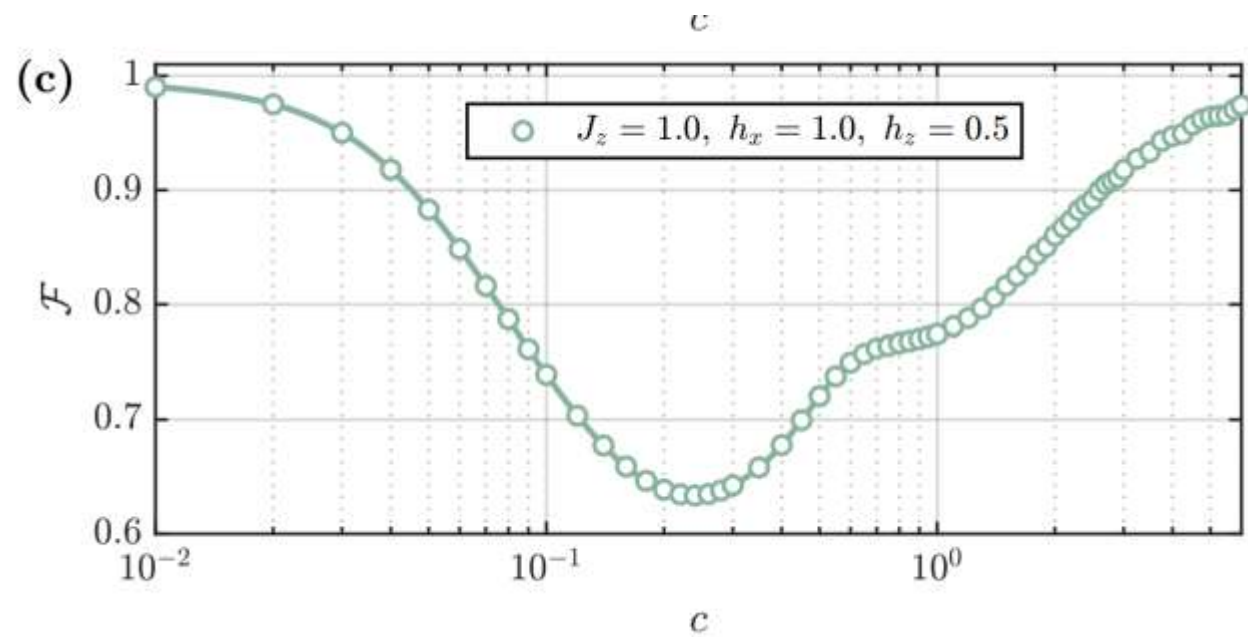
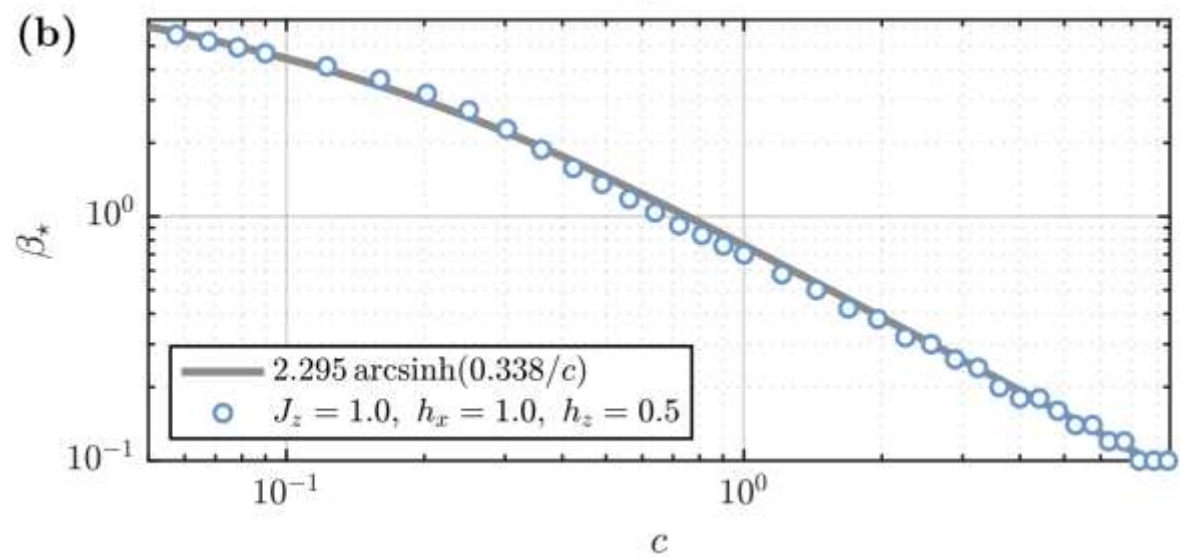
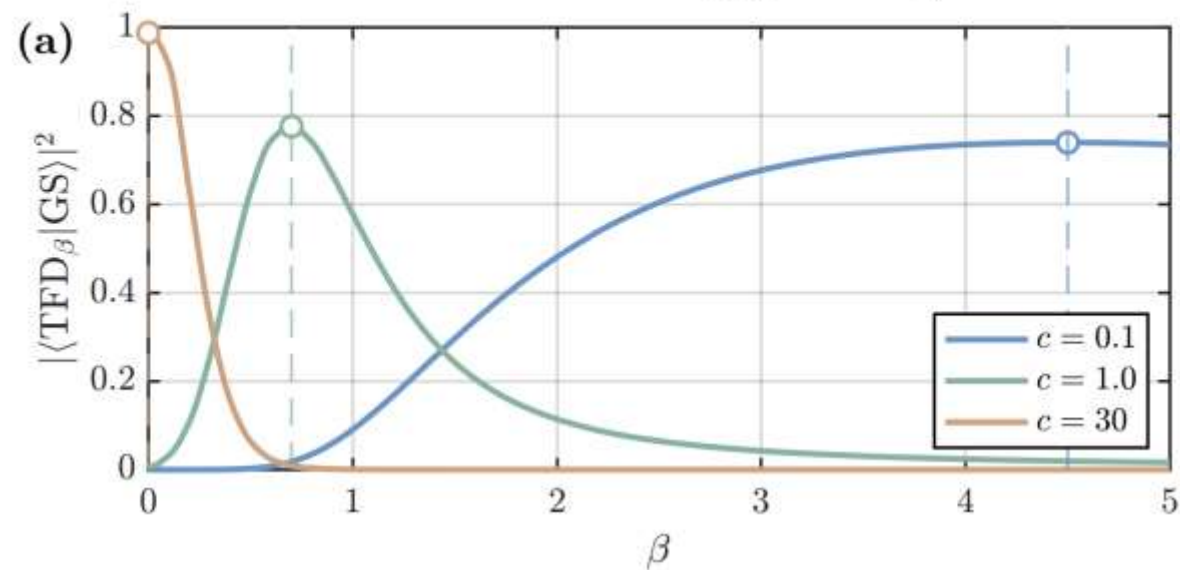
$$+ c \sum_{i=1}^N (X_{i,L} - X_{i,R})^2 + c \sum_{i=1}^N (Z_{i,L} - Z_{i,R})^2,$$

c = LR coupling, need both X and Z couplings



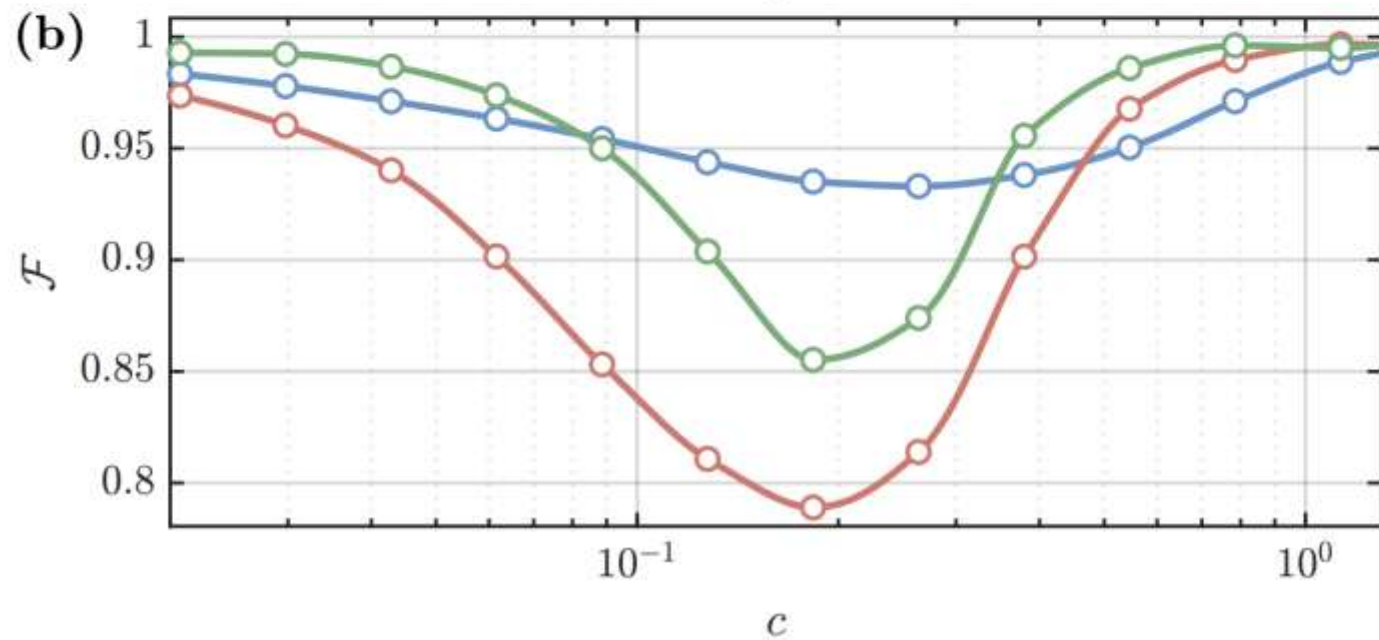
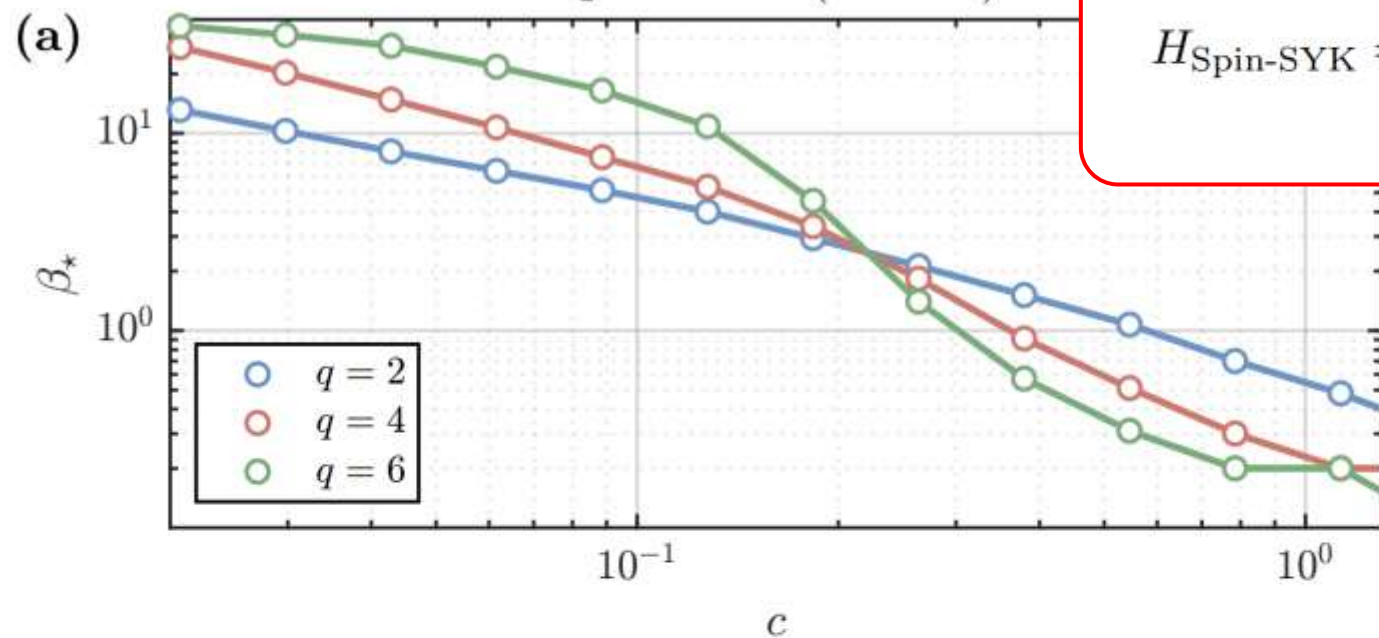
We use MPS tech to compute

Mixed Field Ising ($N = 30$)

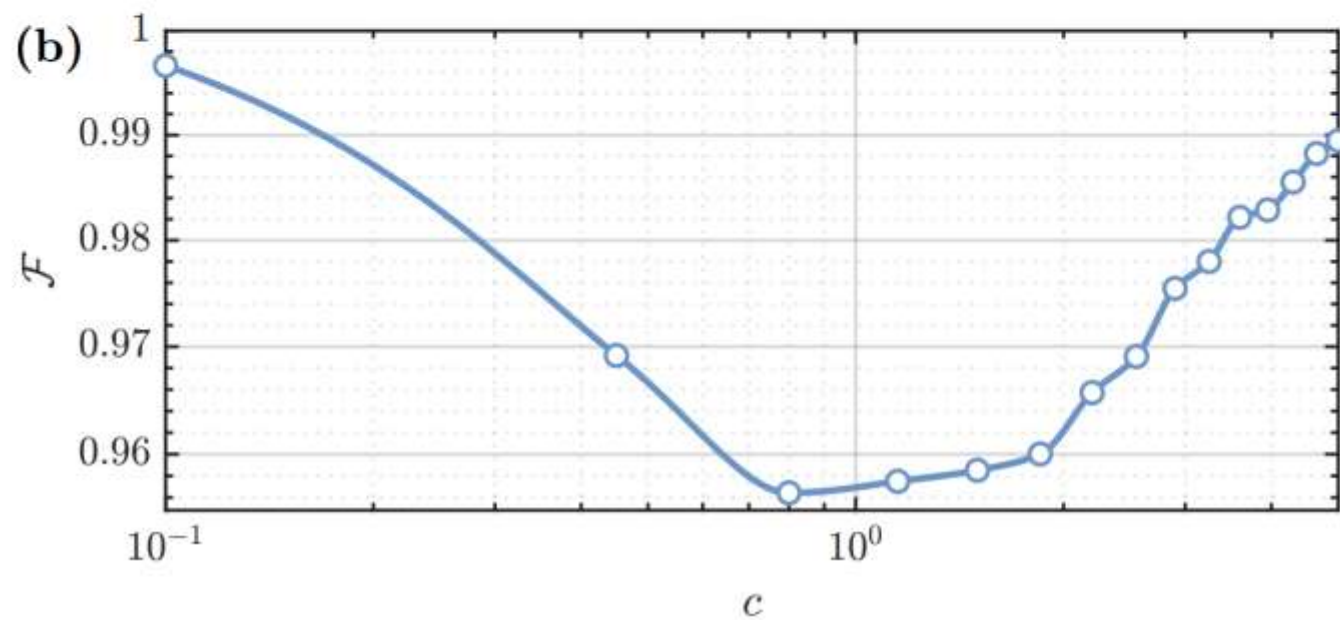
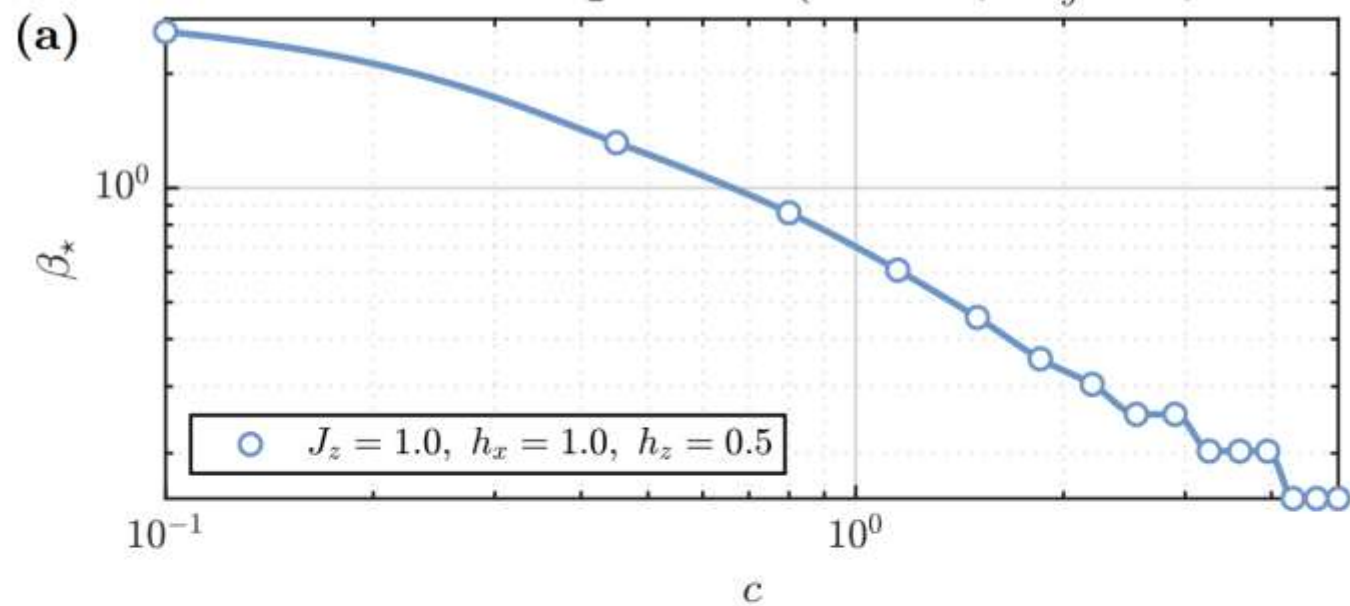


Spin SYK ($N = 8$)

$$H_{\text{Spin-SYK}} = \sum_{i_1 < \dots < i_q} \sum_{\substack{\alpha_1, \dots, \alpha_q \in \{x, y\} \\ \#(\alpha=y) \text{ even}}} J_{i_1 \dots i_q}^{\alpha_1 \dots \alpha_q} \sigma_{i_1}^{\alpha_1} \sigma_{i_2}^{\alpha_2} \dots \sigma_{i_q}^{\alpha_q},$$



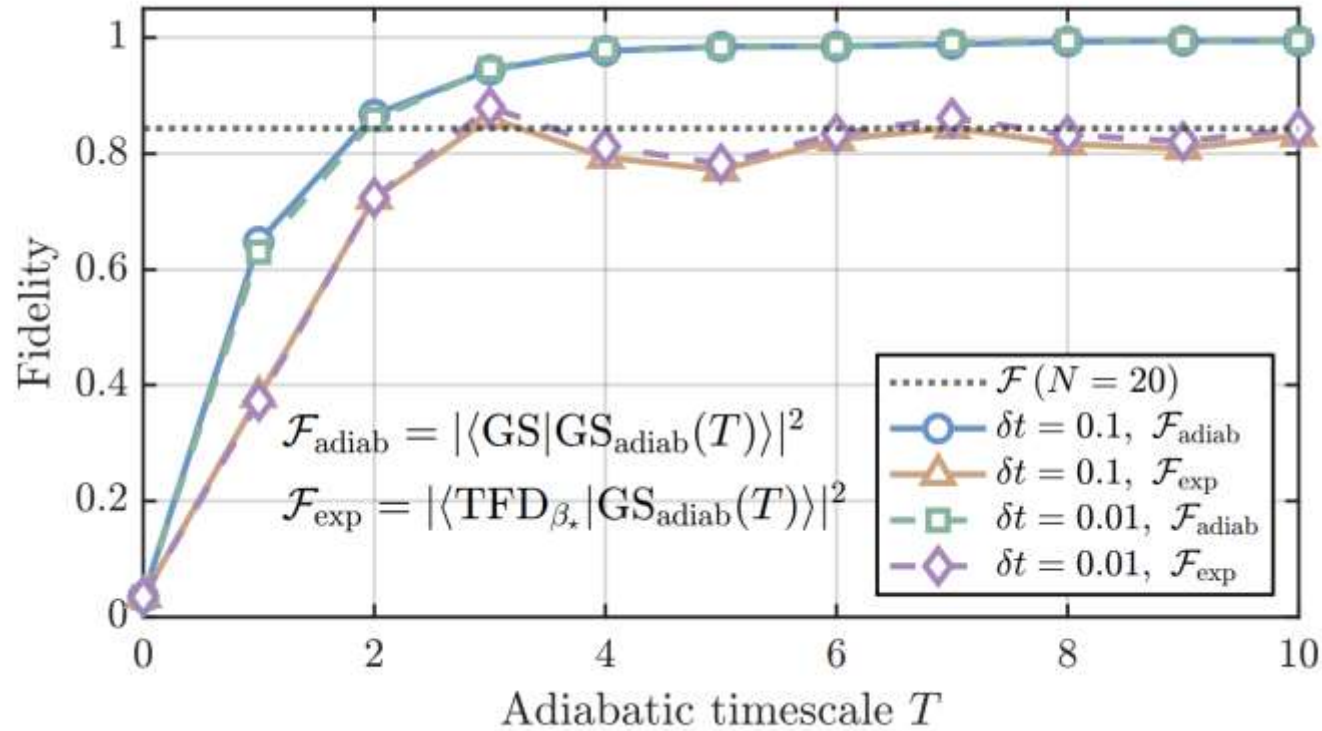
2D Mixed Field Ising Model ($N_x = 2, N_y = 4, N = 8$)



Adiabatic protocol

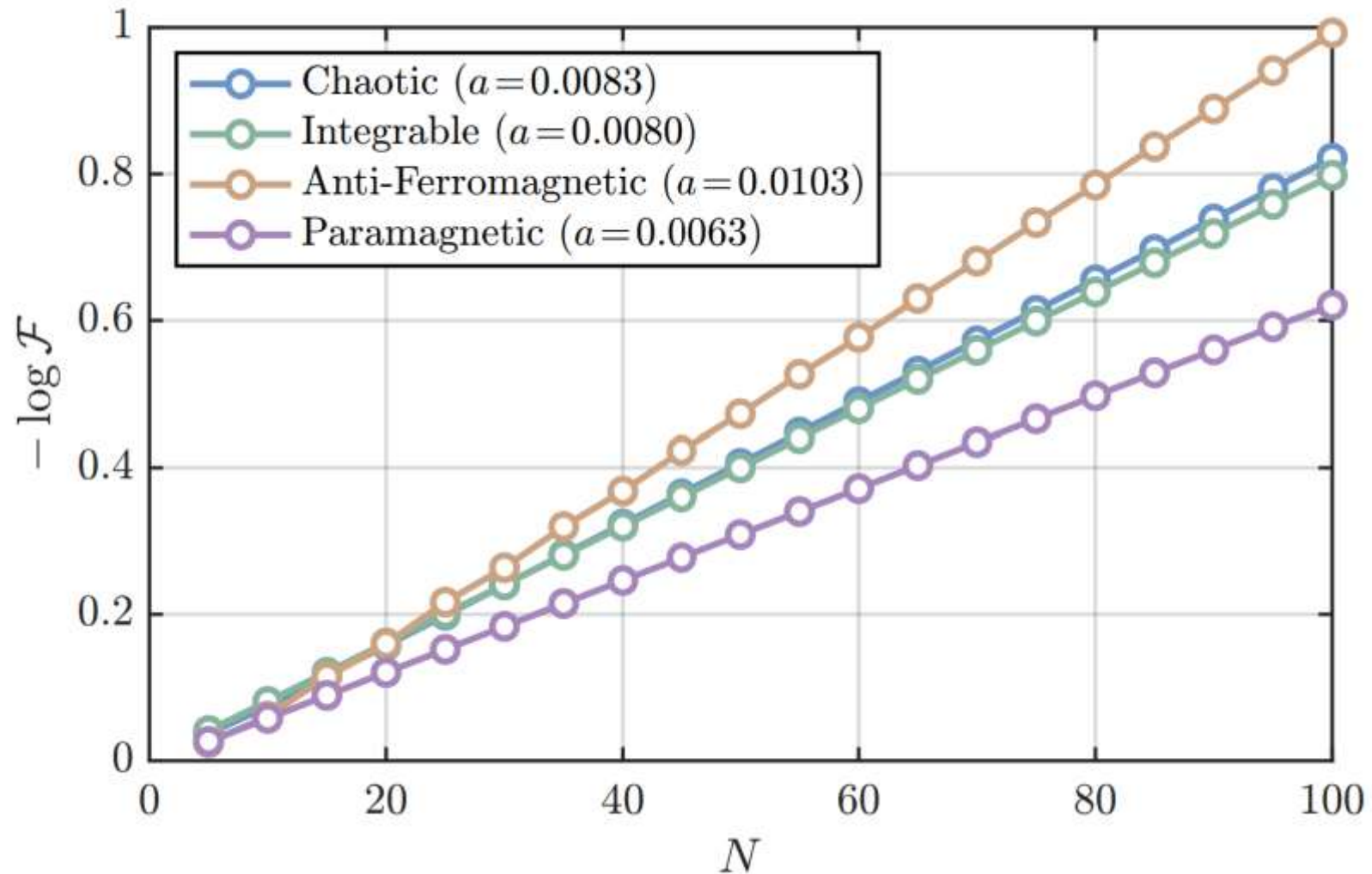
$$H(t) = H_{LR} + s(t)(H_{\text{MFI},L} + H_{\text{MFI},R}^*),$$

$$s(m) = \left(m - \frac{1}{2}\right) \frac{\delta t}{T}, \quad m = 1, \dots, n,$$



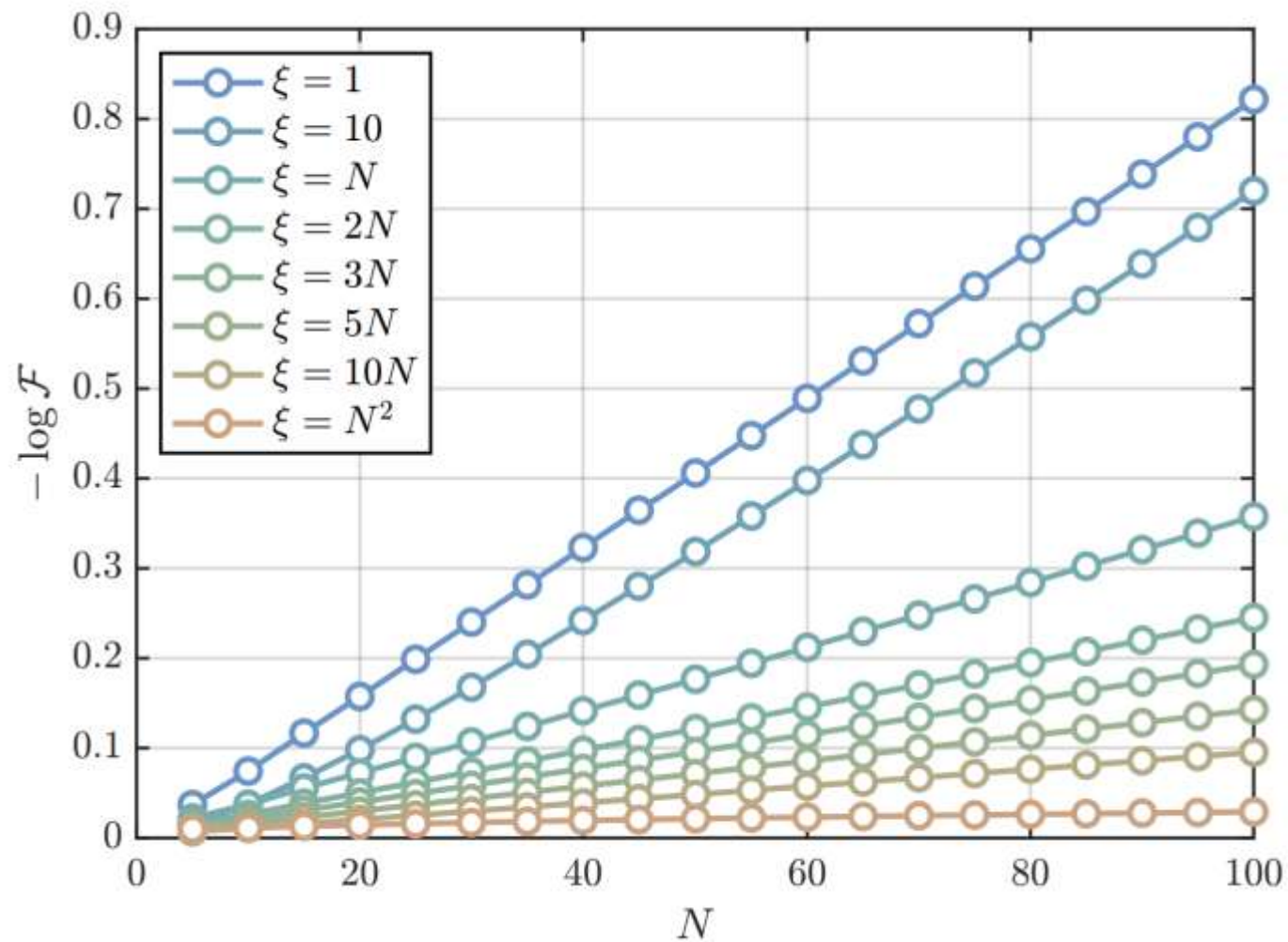
$$\begin{aligned}
 U(s) = & U_{1\text{-body}}\left(s, h_x, h_z, \frac{\delta t}{2}\right) U_{ZZ}\left(s, J_z, \frac{\delta t}{2}\right) \\
 & \times U_{LR}(c, \delta t) U_{ZZ}\left(s, J_z, \frac{\delta t}{2}\right) \\
 & \times U_{1\text{-body}}\left(s, h_x, h_z, \frac{\delta t}{2}\right),
 \end{aligned}$$

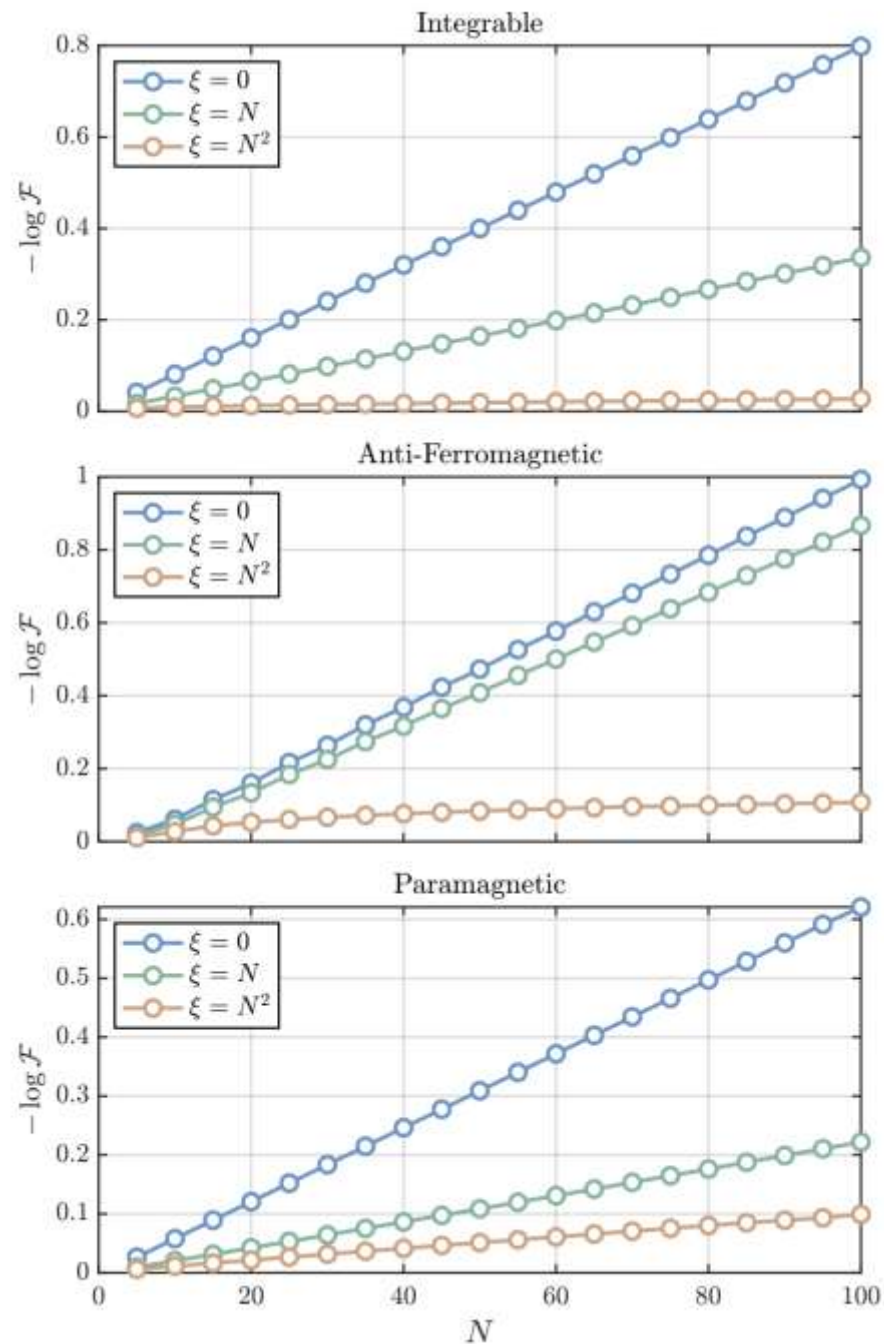
Scaling the system size



Penalty term

$$\frac{\xi}{2N} (H_{0,L} - H_{0,R}^*)^2,$$





Taking $\xi \sim N^a$ is unusual scaling for a term in a natural Hamiltonian, but it presents no obstacle to a poly(N) quantum algorithm

Gravity calculation + general argument

- For the SYK model in its low-energy regime, we can do some further analytics to get (e.g. $\Delta_{\mathcal{O}} = 1/4$):

$$\log \mathcal{F} |_{\xi} \sim -N \times (\text{small coefficient}) \times (N/\xi)^{2-4\Delta_{\mathcal{O}}},$$

$$\mathcal{F} \xi \gtrsim N^{\frac{3-4\Delta_{\mathcal{O}}}{2-4\Delta_{\mathcal{O}}}}.$$

- A general perturbative calculation suggests $\xi \gtrsim N^3$ may be sufficient to stick in the diagonal subspace

$$\|P_{\text{off}}|\text{GS}\rangle\|^2 \sim \frac{N^2}{K} \int d\omega \frac{\hat{\rho}_{\text{off}}(\omega)}{(\Delta_{\text{off}} + \xi\omega^2/N)^2},$$

Outlook

- We presented a candidate general purpose algorithm for TFD prep and gave evidence that it works for a variety of Hamiltonians
- For near term applications, an adiabatic approach prepares a high fidelity TFD state at modest cost
- For devices with logical qubits, one can add a penalty term to prepare large scale high fidelity TFD states
- Can we understand more deeply when and how the algorithm works?

Learning? (preliminary)

- The GS of the LR Hamiltonian is not exactly a TFD of the original Hamiltonian, so what is it?
- Idea: it is the TFD of a “nearby” Hamiltonian
 - $H_{MQ}[H_0] \rightarrow |\text{TFD}[H_0 + \Delta H]\rangle$
 - Adjust H_0 such that $H_0 + \Delta H = H_{\text{target}}$?

Learn a better H_0 by maximizing similarity with local state of a small exact thermal state; we allow H_0 to have all 2-body terms:

(XX, XY, XZ, YX, YY, YZ, ZX, ZY, ZZ, X, Y, Z)

